STEP SEM-3 CodInClub | Python Code Solutions

Category C Coding Assignment Implementation

Problem 1: The Canteen Billing Counter

Task: Compute bill amounts with student/staff discounts and guest service charges [cite: 1].

```
def solve_canteen():
    n = int(input())
    total = 0.0
    for _ in range(n):
        parts = input().split()
        c_type, amt = parts[0], float(parts[1])
        if c_type == "STUDENT": final = amt * 0.90
        elif c_type == "STAFF": final = amt * 0.95
        elif c_type == "GUEST": final = amt + 10.00
        else: final = amt
        total += final
        print(f"{c_type}: {final:.2f}")
    print(f"Total: {total:.2f}")
```

Problem 2: The Campus Parking Charge Calculator

Task: Calculate parking charges for bikes, cars, and trucks based on hourly rules [cite: 1, 2].

```
def solve_parking():
    n = int(input())
    total = 0.0
    for _ in range(n):
        parts = input().split()
        v_type, hrs = parts[0], int(parts[1])
        if v_type == "BIKE": charge = hrs * 10.00
        elif v_type == "CAR": charge = 30.00 if hrs == 1 else 30.00 + (hrs - 1) * 20.00
        elif v_type == "TRUCK": charge = max(100.00, hrs * 50.00)
        else: charge = 0.00
        total += charge
        print(f"{v_type}: {charge:.2f}")
    print(f"Total: {total:.2f}")
```

Problem 3: The Hostel Electricity Bill

Task: Calculate electricity bills for single, shared (split among occupants), and AC rooms [cite: 2, 3].

```
def solve_hostel():
    n = int(input())
    total = 0.0
    for _ in range(n):
        parts = input().split()
        r_type, units = parts[0], int(parts[1])
        if r_type == "SINGLE": bill = units * 8.00
        elif r_type == "SHARED": bill = (units * 6.00) / int(parts[2])
        elif r_type == "AC": bill = (units * 10.00) + 200.00
        else: bill = 0.00
        total += bill
        print(f"{r_type}: {bill:.2f}")
    print(f"Total: {total:.2f}")
```

Problem 4: The Festival Bonus Calculator

Task: Calculate employee festival bonuses for full-time, part-time, and interns [cite: 3, 4].

```
def solve_bonus():
    n = int(input())
    total = 0.0
    for _ in range(n):
        parts = input().split()
        e_type, name, salary = parts[0], parts[1], float(parts[2])
        if e_type == "FULLTIME": bonus = salary * 0.10
        elif e_type == "PARTTIME": bonus = salary * 0.05
        elif e_type == "INTERN": bonus = 2000.00
        else: bonus = 0.00
        total += bonus
        print(f"{name}: {bonus:.2f}")
    print(f"Total Bonus: {total:.2f}")
```

Problem 5: The Streaming Plan Renewal Reminder

Task: Calculate subscription renewal dates using `datetime` and validity periods [cite: 4, 5].

```
from datetime import datetime, timedelta

def solve_streaming():
```

```
    n = int(input())
    for _ in range(n):
        parts = input().split()
        p_type, name, start_date_str = parts[0], parts[1], parts[2]
        days = 30 if p_type == "BASIC" else (90 if p_type == "STANDARD" else 365)
        start_date = datetime.strptime(start_date_str, "%Y-%m-%d")
        renewal_date = start_date + timedelta(days=days)
        print(f"{name}: {renewal_date.strftime('%Y-%m-%d')}")
```